

Week 7 Slides — Speaker Script

Independent Research Project I — From Question to Pre-Analysis Plan

NGN Empirical Finance Research Camp · Prof. Lei Gao (GMU) · Friday, August 7, 2026

Full speaker notes for every slide (also embedded in the .pptx Notes pane). 24 slides.

— Slide 1 — Title — Independent Research Project I

Welcome to Week 7 — this is the session you have been climbing toward all camp. Before we start, drop your Week 6 deliverable in the chat: the AI literature map plus your verification log. Got it? Good. Now here is what today is. For six weeks you have been a consumer of research. You learned probability, then regression, then two full weeks of causal design, and then you learned to read the frontier — first by hand, then with AI. Today you stop consuming and start producing. By two o'clock, every one of you walks out with a one-page design for your own study. Not results — we are not running anything today. A design. And here is the bar we hold it to: a plan honest enough that even a null result, a finding of nothing, would still teach us something. That is what separates a real study from fishing. Let's go build yours.

— Slide 2 — The 8-week map — where we are

Let's find ourselves on the map. Weeks one through four built your toolkit — probability, the OLS engine, and two full weeks of causal design. Weeks five and six taught you to read the frontier, first by hand and then with AI as an accelerant. Every one of those weeks was about consuming what other people made. Today, Week 7, the arrow flips. This is Research Project One, where you take the topic and the literature map you have been building and turn them into your own empirical design. Next week, Week 8, you execute it. So notice the tagline on the slide: you have read the frontier — now you plant your flag on it. Everything you have learned was preparation for exactly this moment. The skills weeks are behind you. From here on you are the researcher, and the work on the screen is your work, not somebody else's.

— Slide 3 — The pivot — consume to produce

Here is the whole pivot in one picture. On the left, six weeks of input. You consumed — the toolkit, the paper teardown, the verified literature map. That was you learning to judge other people's research. Today, in the middle, the arrow turns: you produce. You take all of that judgment and point it at a question of your own. And on the right is the bar, the single standard I want you to hold your design to all day: it has to be honest enough that a null result — finding nothing — would still be informative. Think about what that means. If the only way your project is interesting is if you get a big, significant effect, then you are not doing science, you are hoping. A good design tells us something whether the effect is there or not. That is the target. Everything in the next two hours is about getting your project to clear that bar.

— Slide 4 — Block A — From a Question to a Design

Alright, Block A. Before you can plan a study, you have to turn a vague interest into a sharp, answerable question — and then into a design. This is where most projects quietly live or die, and it happens in the first ten minutes of thinking, not the last. I am going to give you four tools.

A filter that kills weak questions in ninety seconds. A two-by-two that tells you which surviving questions are actually worth your two weeks. A map that takes your question down to a real dataset. And then the single most important sentence you will write all camp — your identifying assumption. The goal is not a perfect question. It is a researchable one you can actually build in the time you have. Let's start.

— Slide 5 — The so-what filter — three questions

First tool: the so-what filter. Most research ideas are bad, and that is fine — the skill is killing the bad ones fast so you can spend your energy on a good one. Run every idea through three questions. One: is the answer already known? If a quick search shows five papers already nailed it, move on. Two: would anyone actually change a decision if they knew the answer? If nobody — no investor, regulator, manager, nobody — would do anything differently, it is trivia, not research. Three: can you actually measure both the cause and the effect? A beautiful question about something you cannot observe is a dead end. An idea that survives all three questions is worth your two weeks. An idea that fails any one of them, you drop right now, in ninety seconds, with no regret. Be ruthless here — it is the cheapest place in the whole project to say no.

— Slide 6 — Feasibility × novelty — the 2×2

Second tool: the feasibility-by-novelty two-by-two. Put novelty on one axis — is this a fresh question or a done question — and feasibility on the other — can you actually get the data. That gives four boxes. Top right is where you want to live: feasible and fresh. A new question you can actually answer with data you can actually get. Top left is been-done: clean, easy data, but no new question — you would just be re-running someone's paper. Bottom right is the trap that catches ambitious students: heroic data. A gorgeous, important question that needs data no undergraduate can get in two weeks. And the bottom left you just skip. Your job today is to move your idea into that top-right box. Sometimes that means shrinking a heroic question down to a feasible slice of it. A smaller question you can actually answer beats a grand one you cannot.

— Slide 7 — The cast → dataset map

Third tool: the cast-to-dataset map. Take your question and pull out three things — who or what is it about, what is the outcome you care about, and when does the action happen. Then, for each one, name the actual dataset. Who: firms, individuals, municipalities — that is your unit and your identifier. What: the outcome variable, and which source has it. When: the event or the timing, and where those dates live. Watch how this works on Rainbow. Who: municipal bond issuers. What: their borrowing costs, from bond pricing data. When: the dates states adopted same-sex-marriage bans, from legislative and court records. Here is the hard rule that makes this tool bite: if you cannot name the dataset — the actual file, the actual source — you do not yet have a project. You have a wish. Naming the data is what turns a topic into something you can build.

— Slide 8 — The identifying-assumption sentence

Now the most important sentence you will write all camp — your identifying assumption. Every credible empirical paper has one, stated or not, and today you make yours explicit. Here is the template on the screen, and I want you to fill in every blank. Comparing my treated units to my

controls, my estimate of the effect is causal if this assumption holds — which is plausible because of this design feature. Read that again. It names who you compare, what you are estimating, the one assumption the whole causal claim rests on, and the concrete reason a skeptic should believe that assumption. If you cannot fill in the blank after because, you do not have identification — you have a correlation with a hopeful label. This sentence is not decoration. It is the load-bearing wall of your entire project. Everything else — the data, the regression, the robustness checks — exists to defend this one sentence.

— Slide 9 — Worked live — the Rainbow identifying assumption

Let's fill in that sentence live, on Rainbow of Municipal Credit. Watch each blank. Comparing states that adopted a same-sex-marriage ban to states that had not yet adopted one, our estimate of the effect of the ban on municipal borrowing costs is causal if the two groups' borrowing costs would have moved in parallel absent the ban — which is plausible because the timing of adoption was driven by court and legislative calendars, not by a state's credit trajectory. Notice what the because clause is doing. The whole worry with a policy study is that states adopt the policy precisely when their economy is already changing. The defense here is that the ban timing was set by unrelated legal and political calendars, so it is not lining up with credit conditions. That is a real, checkable argument — not a hope. This is exactly the sentence each of you will write for your own project by two o'clock.

— Slide 10 — The threats-and-responses table

Now the threats-and-responses table, and I want you to take this one seriously, because this table is literally the first page a referee reads. For every threat to your causal story, you write the design response — how your design already handles it. Confounding trend: some other shock hits your treated group at the same time. Response: check pre-trends and control for time-varying covariates. Reverse causality: maybe the outcome drives the treatment, not the other way around. Response, on Rainbow: the timing is set by legal calendars, not credit. Selection: treated and control groups differ systematically. Response: difference-in-differences differences out fixed group differences, and you use not-yet-treated units as controls. Measurement error: your outcome is noisy. Response: use the cleanest source you have. Here is the point — when you build this table yourself, before anyone else does, you are refereeing your own paper. You catch the objections while they are still cheap to fix.

— Slide 11 — Block B — The Pre-Analysis Plan & the Data

Block B. You have a sharp question, an identifying assumption, and a threats table. Now we do two things that turn that design into something real and defensible. First, the pre-analysis plan: you write down exactly what you are going to test, and how, before you look at a single outcome. That is what makes a null result believable and an effect trustworthy. Second, the data: how to pick the right source and pull it so cleanly that anyone — including you, six months from now — could reproduce every number. By the end of this block you will have the recipe for both the plan and the pull. This is the unglamorous machinery that separates a study people believe from a story people wave away. Let's get into it.

— Slide 12 — Why pre-register — the garden of forking paths

Why bother pre-registering — writing down your plan before you see the data? Because of the garden of forking paths. Here is the trap. When you analyze data, you make dozens of small

— Slide 13 — The Olken short-form PAP — six items

So what actually goes in a pre-analysis plan? You do not need a fifty-page document — you need Olken's short form, six items, one to two pages total. One: directional hypotheses. Not just there is an effect, but which sign you expect and why. Two: your one primary specification — the single regression that is the headline test. Three: secondary and heterogeneity analyses — the follow-up questions, clearly labeled as secondary. Four: a multiple-testing correction. If you test several outcomes, you use Benjamini-Hochberg to control the false-discovery rate, so you do not get to celebrate the one lucky star out of twenty. Five: falsification and placebo tests — checks that should come back empty if your design is clean. Six: what would falsify your main hypothesis — state, in advance, the result that would prove you wrong. That last one is the tell of a serious researcher. The value here is commitment, not length. Keep it tight, and file it on time.

— Slide 14 — The primary specification — name every part

Let's build the piece that everything hinges on: the primary specification. The rule is that you can name every single part out loud. Look at the Rainbow equation on the screen. The outcome is the municipal offering spread — what it costs the issuer to borrow. The treatment is the ban indicator, switching on after a state adopts. The controls are issue and issuer characteristics. The fixed effects are issuer and year, soaking up fixed differences and common time shocks. The standard errors are clustered by state, because that is the level the policy varies at. And the estimator is a staggered difference-in-differences done properly — the Callaway and Sant'Anna approach from 2021, which handles the fact that states adopt at different times without the bias a naive two-way regression can produce. Every part has a name and a reason. When you can recite yours like this, from outcome to estimator, you have a specification — not a vague intention to run some regressions.

— Slide 15 — Falsification / placebo tests

Falsification and placebo tests — this is the move that turns a skeptic into a believer, so design at least one. The idea is beautifully simple: build a test that should return nothing if your design is clean. If it returns nothing, great, your story survives a real attempt to break it. If it lights up, you just caught your own problem before a referee did. Three flavors. Pre-trend test: check that treated and control groups were moving in parallel before the treatment — if they were already diverging, your parallel-trends assumption is in trouble. Placebo date: pretend the policy happened two years earlier than it did and re-run it — you should find no effect at the fake date. Unaffected group: run the same design on a group the policy should not have touched — again, you want nothing. A design that passes tests that were genuinely capable of failing is a design people believe.

Slide 16 — Data acquisition — pick the right source

Now the data — where do you actually get it? Pick the right source for your question. WRDS is your gateway to CRSP for stock prices and returns, and Compustat for accounting fundamentals — the workhorses of finance. EDGAR gives you the raw SEC filings themselves. FRED is the home of macro and interest-rate series. HMDA is loan-level mortgage data — that is exactly the source behind the Gao and Sun fair-lending work in PNAS. Thirteen-F filings give you institutional holdings. PatentsView is your source for innovation and patent data. And yfinance is a quick, free way to pull market prices when you do not need the heavy machinery. The skill is not knowing all seven cold — it is matching your outcome variable to the one source that actually has it, cleanly, at the frequency you need. Name your source before you write a line of code.

Slide 17 — Pull reproducibly — four habits

Getting the data is not enough — you have to pull it reproducibly, so that every number in your paper can be regenerated from scratch. Four habits. One: cache-or-fetch. Your code checks for a local copy first and only downloads if it is missing, so you are not hammering the server and your results do not change under you. Two: pin a data vintage. Databases get revised — write down exactly which version, which download date, you used, so your numbers are frozen in time. Three: keep an audit log. Record what you pulled, when, and how, so future-you can retrace every step. Four, and this is the big one: never hand-edit a downloaded file. The moment you open a spreadsheet and fix a value by hand, your pipeline is broken and no one can reproduce you. Every change happens in code. If you cannot rerun your whole project from raw data with one command, it is not reproducible — and reproducible is the standard.

Slide 18 — Building the analysis dataset — four traps

Now you build the analysis dataset, and this is where silent, deadly bugs live. Four traps. First, identifier crosswalks. Firms carry different IDs in different databases — PERMNO in CRSP, GVKEY in Compustat, CIK in EDGAR — and you have to join on the right crosswalk or you will silently merge the wrong companies. Second, survivorship and look-ahead bias. If you only keep firms that survived to today, or you accidentally use information that was not known yet at the time, you bake the answer in. The fix is an as-of merge — join on what was actually knowable at each date, not a naive join. Third, winsorizing. Outliers can hijack a regression, so trim them by a stated rule — say, at the first and ninety-ninth percentile — and write the rule down. Fourth, missing data: handle it on purpose, with a stated choice, not by silently dropping rows and hoping. Clean data, built deliberately, is half the paper.

Slide 19 — The one rule — file the PAP as a tagged commit

Here is the one rule from all of Block B I most want you to carry out the door. File your pre-analysis plan as a tagged commit before you run your first outcome regression. Let me say why that specific mechanic matters. A tagged commit in your repository carries a timestamp that you cannot fake after the fact. So when you later report your result, the timestamp is your evidence that you committed to the plan before you saw the answer — that you did not fish, did not fork, did not tune the specification to the result. It is the difference between saying trust me, I planned this and being able to prove it. Two minutes of work — write the plan, commit it, tag it — and you have converted your honesty into something checkable. Do it before the first

— Slide 20 — Block C — Workshop: Lock Your Own Project

Block C — the workshop. This is the payoff, so we protect the time. You have the so-what filter, the identifying-assumption sentence, the threats table, the pre-analysis plan, and the reproducible data pull. Now you stop watching me do it on Rainbow and you do it on your own project. In the next half hour you are going to draft one page for your study — the assumption sentence, a threats table, and your primary specification line — and then you are going to read it out loud and let the room stress-test it. This is the moment a generic design becomes your design, not a copy of mine on Rainbow. Two crisp minutes each, and the room's whole job is to find the threat you couldn't see from the inside. Keep it moving, and let's lock your project in.

— Slide 21 — Your one page — three things

Here is your one page — three things, and only three, for your own project. Part A: the identifying-assumption sentence. Fill in every blank from slide eight — treated, controls, effect, the assumption, and the because. Part B: a three-row threats-and-responses table. Pick your three biggest threats and write the design response to each. Part C: the primary-specification line. Outcome, tilde, treatment, plus controls, plus fixed effects, clustered by something, estimated by something — every part named, like we did for Rainbow. That is the whole deliverable for the next twenty minutes. One page. If you get all three of these down honestly, you have the design half of your capstone paper, and everything in Week 8 is just execution. Do not polish the prose. Get the logic right. Start now, and I will come around.

— Slide 22 — Round-robin — read it aloud, name a threat

Round-robin — your turn, out loud, and this is the single most valuable two minutes you will spend today. Each of you reads your identifying-assumption sentence to the room. Just that one sentence — who you compare, what you estimate, the assumption, and the because. Then the room does its job: everyone else names one threat you missed. Not to tear you down — to save you. Catching a fatal flaw here, in this room, in August, is free. Catching it after you have pulled the data and run the regressions is not — that is a week of your life. So when it is your turn, say it clearly and then actually listen. And when it is someone else's turn, be the person who spots the confounder they did not see. The best thing that can happen to you today is that a classmate finds the hole in your design while it is still cheap to fix.

— Slide 23 — Wrap & assignment — the Pre-Analysis Plan + Identification Memo

Let's wrap and set the assignment. This week you write the Pre-Analysis Plan plus Identification Memo for your own project — the identifying assumption, the threats table, the primary specification, the falsification tests, and a reproducible data-pull plan. Keep it tight: one to two pages. The value is commitment, not length. This is due before the Week 8 session, Friday, August 14. And understand what it is: this is the design half of your capstone paper, the one due August 15. Week 8 is where you take this exact plan and execute it — pull the data, run the primary specification, run the falsification tests, and write up whatever you find, effect or null. So the more honest and complete your plan is this week, the smoother next week goes. Everything has been pointing here. Go plant your flag, commit your plan before you peek, and bring me a design you would defend to a skeptic.

Slide 24 — Thank you & references

Let me close with the line that runs through this whole session. A good design is one you would defend to a skeptic — and one you filed before you saw the answer. That is the standard, whether your result comes back as a big effect or as a flat null. The tools on the honest references list are the ones behind everything we did today. Olken's 2015 piece in the Journal of Economic Perspectives on the promises and perils of pre-analysis plans. Callaway and Sant'Anna's 2021 method for difference-in-differences with staggered timing. And the running anchors from Prof. Gao's own work — the Rainbow of Municipal Credit, Gao and Sun's 2019 fair-lending study in PNAS, and Deng, Gao and Kim's 2020 Reg SHO natural experiment in the Journal of Corporate Finance. Bring your Pre-Analysis Plan and Identification Memo next Friday, August 14, and come ready to execute. You have read the frontier. Now go plant your flag. See you next week.